

# JECRC UNIVERSITY JAIPUR

Session: 2025–26

**Subject:** Mathematics-I (DMA024A)

**Course:** B. Tech - I Semester

## Tutorial Sheet – 8

UNIT – 3

M.M: 64

**CO3:** The tool of power series and Fourier series for learning advanced Engineering Mathematics.

### PART – A (5 × 2 = 10 marks)

1. Define D'Alembert's Ratio test.
2. Define Raabe's test.
3. Define Cauchy's Root test.
4. Test the convergent of the series  $\sum_{n=1}^{\infty} \frac{n}{n+1}$ .
5. Test the convergence of the series  $\sum \frac{1}{n!}$ .

### PART – B (3 × 7 = 21 marks)

1. Test the convergence of the series  $\sum \frac{n^n}{(n+1)^{n^2}}$
2. Test the convergence of the series  $\sum \frac{2n-1}{n(n+1)(n+2)}$
3. Test the convergence of the series  $\sum \frac{(4n-3)^2}{(4n)^2}$

### PART – C (3 × 11 = 33 marks)

1. Test the convergence of the series  $1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \dots$
2. Test the convergence of the series  $1 + \frac{1}{2}x + \frac{2!}{3^2}x^2 + \frac{3!}{4^3}x^3 + \dots$
3. Test the convergence of the series  $\sum \frac{1.3.5\dots(2n-1)}{2.4.6\dots 2n} \frac{1}{n}$ .

### Answer Key:

#### Part A

4. Not converge, 5. Converge

#### Part B

1. Convergent, 2. Convergent, 3. Divergent

#### Part C

1. Converge  $x \leq 1$  and diverge  $x > 1$ , 2. Converge  $x < e$  and diverge  $x \geq e$ , 3. Convergent